INVERTIS UNIVERSITY, BAREILLY

FACULTY OF ENGINEERING & TECHNOLOGY

END-SEMESTER EXAMINATION 2022-23

Academic Session: 2022-2023

Programme: B.Tech. (Computer Science & Engineering) **Semester: Semester III**

Course Code: BCS-303 **Course Title: Digital Electronics**

Time Allowed: 3 Hours Maximum Marks: 50

Note: Attempt all sections as per instructions. Assume missing data suitably, if any.

SECTION A (Short Questions - 2 Marks each)

Q1(a). State De Morgan's Theorems with Boolean algebraic proofs.

Q1(b). Convert hexadecimal (2F9A)16 into octal and binary equivalents.

Q1(c). Define race-around condition in JK Flip-Flop and how to avoid it.

Q1(d). Distinguish between Combinational and Sequential circuits.

Q1(e). What is a Multiplexer (MUX)? Give its application.

SECTION B (5 Marks each)

Q2. Minimize the Boolean function using K-Map: F(A,B,C,D) = Sum m(0,1,2,5,7,8,9,10,13,15).

Q3. Design a 4-bit synchronous binary up/down counter using T flip-flops.

Q4. Implement a Full Adder using two 4:1 Multiplexers.

SECTION C (10 Marks each)

Q5. Differentiate between RAM, ROM, PLA, and PAL with block diagrams and programming tables.

Q6. Explain Master-Slave JK Flip Flop with circuit diagram, truth table, and timing waveforms.

========== END OF QUESTION PAPER ==========